SECRET AGENT

Case File

Crack it. Check it. Beat the clock.



AGENT ID

Name: _____

Partner: _____

Event: Codebusters (Field A Fridays)

If found, please return to the DDMES Science Olympiad binder shelf.

How a codebuster works

Welcome, Agent. Spies, soldiers, and computers all hide messages in **ciphers**. Your job: learn the cipher types on the test, crack a case at home after every meeting, and practice until you and your partner can beat the clock.

1
STUDY
Learn the keys on the reference pages (p. 4-5).

2
CRACK
After each meeting, open that meeting's case (p. 6-10).

3
CHECK
Turn the book over for the answer key on the last page.

4
RACE
Time your memory keys (p. 11). Log practice (p. 13).

5
EARN
Show a coach, get initials, color the badge.

The Agent's Code

- [ ] Work the case before you peek at the answers.
- [ ] Check your work: 2 small mistakes are OK, 3 cost points.
- [ ] Share the work with your partner. Take turns.
- [ ] Write neatly so your partner can read it.
- [ ] Keep secret messages kind. Agents never tease.
- [ ] Never share real passwords, only pretend ones!

My field kit

- [ ] This case file and a sharp pencil
- [ ] An eraser
- [ ] Index cards for key flashcards
- [ ] A timer (phone or kitchen timer, ask an adult)
- [ ] Your partner or a family member

Journal map

Know your event	p. 3
Reference pages	p. 4-5
Cases 1-5	p. 6-10
Memory race	p. 11
Cipher swap and log	p. 12-13
Missions, badges, key	p. 14-16

Where to look

Cereal boxes and books
count letters to see which ones are most common

Your knuckles
tap Tap Code messages on a table

Puzzle books
cryptograms are Aristocrats!

Family notes
leave a cipher on the fridge for someone

Car rides
say the alphabet backward (Atbash practice)

Clocks and wheels
a Caesar shift is like turning a wheel

Did you know? In the U.S. Civil War, soldiers sent secret messages with a cipher disk, and in World War II code talkers helped keep messages safe!

Know your event: Codebusters

Big idea: Spies, soldiers, and computers all hide messages in codes. In Codebusters you learn ten kinds of ciphers and the tricks for cracking them, then race your partner against the clock.
Tournament format: A 50-minute written test with only pencils: partners decode as many secret messages as they can, using a resource sheet (letter frequencies, Atbash, Galactic Alphabet, Baconian and Vigenere tables) but NO PigPen, Tap Code, or Knight's Templar keys, so those must be memorized.

Given on the resource sheet

Letter frequencies, Atbash, Galactic Alphabet, Baconian table, and Vigenere tables.

NOT given: memorize these!

- [ ] PigPen key
- [ ] Tap Code grid
- [ ] Knight's Templar key

How scoring works

2 or fewer mistakes on a question still gets **full points**. Each extra mistake loses **50**. Example: a 200-point question.

Mistakes	Points
0, 1, or 2	200 (full points!)
3	150 (200 - 50)
4	100 (200 - 100)
5	50 (200 - 150)

The cipher types on the test

Cipher	What it is	Practice
Caesar (shift)	Every letter moves 1, 2, or 3 places.	p. 4, 6
Atbash	The alphabet flips: A=Z, B=Y.	p. 4, 6
Tap Code	Row and column numbers.	p. 5, 7
PigPen	Grid and X shapes, some with dots.	p. 5, 7
Aristocrat	Every letter swapped. Solve with clues.	p. 8, 9
Vigenere	A key word gives a new shift for each letter.	p. 4, 9
Baconian	5 symbols of 2 kinds = 1 letter.	p. 5, 10
Knight's Templar	Symbol key you must memorize.	p. 5, 11
Galactic Alphabet	Strange symbols. Build the key from a sentence.	resource sheet

Partner plan

- First minute: look over the test and split the questions.
- Draw the 3 memory keys on scrap paper first.
- Do the fastest questions first. Most teams don't finish every question.

Agent tips

- Made by moving forward? Decode by moving back.
- After Z comes A again.
- A real sentence means you got it right.

Code book: shifts, mirrors, and letters

To **encode** with a shift, find the real letter on top and go down to your shift. To **decode**, find the cipher letter in the shift row and go up.

Real letter	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Shift +1	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
Shift +2	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
Shift +3	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
Shift -1	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y
Shift -2	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Shift -3	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Atbash	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A

How to crack a shift

1. Find the shortest word.
2. Try moving its letters back 1, 2, or 3, then forward 1, 2, or 3.
3. When it turns into a real word, you found the shift.
4. Use the same shift on every letter.

Atbash trick

Write A to M, then Z to N under it. The same key encodes **and** decodes.

A B C D E F G H I J K L M
Z Y X W V U T S R Q P O N

Letter detective tricks (for Aristocrats)

Most common letters	E T A O I N S H R (E is the most common).
Short words	One letter: A or I. Common: THE, AND, IS, IT, TO, OF.
Word pattern	Letters that repeat, like the two L's in BELL (pattern 1-2-3-3).
Golden rule	No letter ever stands for itself.

Vigenere letter numbers

Letter	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Number	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

How to decrypt a Vigenere

1. Write the key word under the ciphertext again and again (skip spaces).
2. Change both letters to numbers.
3. Cipher number minus key number. Below 0? Add 26.
4. Change the answer back to a letter.

Speed drill: Z to A

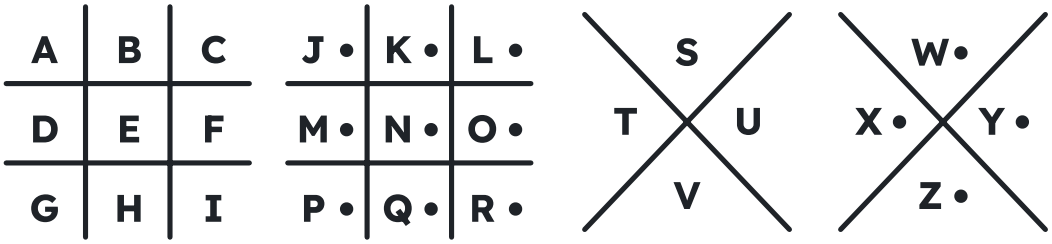
Cover this page. Write the alphabet backward from memory. Time yourself!

Try 1: ___ sec Try 2: ___ sec Try 3: ___ sec

Code book: keys to memorize

These 3 keys are NOT on the resource sheet. Draw them from memory on scrap paper as soon as the test starts.

1. PigPen



Each letter's symbol is the shape of the lines around its space, plus the dot if it has one. Examples:



2. Tap Code

	1	2	3	4	5
1	A	B	C/K	D	E
2	F	G	H	I	J
3	L	M	N	O	P
4	Q	R	S	T	U
5	V	W	X	Y	Z

The grid is 5 by 5. Rows: ABCDE / FGHIJ / LMNOP / QRSTU / VWXYZ. C and K share one box, so there is no K. First number = row (down), second = column (across): 2 3 = H.

3. Knight's Templar

Copy the key from your Templar key sheet (Lesson 4)

Ask a coach to check your copy. Then practice it from memory on p. 11.

Baconian table (on the resource sheet)

24 letters: I and J share, U and V share. Sort the symbols into an A group and a B group, then look up each group of 5.

Letter	Code	Letter	Code
A	AAAAA	N	ABBAA
B	AAAAB	O	ABBAB
C	AAABA	P	ABBBA
D	AAABB	Q	ABBBB
E	AABAA	R	BAAAA
F	AABAB	S	BAAAB
G	AABBA	T	BAABA
H	AABBB	U/V	BAABB
I/J	ABAAA	W	BABAA
K	ABAAB	X	BABAB
L	ABABA	Y	BABBA
M	ABABB	Z	BABBB

Case 1: Operation Shift and Mirror

TOP SECRET

Open this case: after Lesson 1 (Fri Oct 2) · **Badge:** Shift Cracker

> **Your mission:** headquarters intercepted 5 messages. Crack them with the shift table and the Atbash trick on page 4. Write the real letter on the line under each cipher letter.

1. Caesar (Hint: it was made with shift +1.)

U I F Q B T T X P S E J T Q B O D B L F

__ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __

2. Caesar (Find the shift yourself.)

Y Q L Y G J A Y L Q J C C N D M P R F P C C

__ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __

W C Y P Q

__ __ __ __ __

3. Caesar (Find the shift yourself.)

K L G H W K H P D S X Q G H U W K H E L J U R F N

__ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __

4. Atbash

U I L T H W I R M P D Z G V I G S I L F T S

__ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __

G S V R I H P R M

__ __ __ __ __ __ __ __ __

5. Atbash

G S V X Z P V R H R M G S V U I V V A V I

__ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __

Shifts I found

Message 2: shift _____ Message 3: shift _____

Remember: if the message was made by moving forward, you decode by moving back.

Your turn: write a short message in Atbash for someone at home

Case 2: Operation Knock-Knock TOP SECRET

Open this case: after Lesson 2 (Fri Nov 6) · **Badge:** Tap Talker and PigPen Pro

> **Your mission:** draw the Tap Code grid and the PigPen key from memory in the boxes first (no peeking at p. 5!). Then crack the messages.

Tap Code grid from memory

PigPen key from memory

Tap Code messages

Each pair is row, then column. A wide space means a new word.

1. Tap message

32 15 15 44 11 44 44 23 15 22 11 44 15

___ ___ ___ ___ ___ ___ ___ ___ ___ ___ ___ ___ ___

2. Tap message

52 15 14 24 14 24 44

___ ___ ___ ___ ___ ___ ___

Knock-knock challenge: tap a 3-letter word to someone at home with your knuckles. Tap the row, pause, tap the column. Did they get it? Word I tapped: _____ They guessed: _____

PigPen messages

3. PigPen message



___ ___ ___ ___ ___ ___ ___ ___ ___

4. PigPen message



___ ___ ___ ___ ___ ___ ___ ___ ___ ___ ___ ___

Case 3: Operation Letter Count TOP SECRET

Open this case: after Lesson 3 (Fri Dec 11) · **Badge:** Letter Detective

> **Your mission:** this message is an **Aristocrat**: every letter was swapped for a different letter, and no letter stands for itself. **Hint: the first word is THE.** Write T, H, and E over every matching cipher letter, then use the detective tricks on page 4.

1. Aristocrat (200 points)

R T H F H D R D G Z H D J L H O A A I J R

__ __

X Z D R H Q Z Q O J Q I R T H E Q A R Z W H R T H

__ __

D P J X X R T Z Q O D

__ __ __ __ __ __ __ __ __ __ __ __

Count the top cipher letters

Cipher letter	How many times	I think it is
H		
R		
D		
Q		
Z		

Letter hunt at home

Pick a page from a book or cereal box. Count the first 100 letters. Tally each letter below. Is E the winner?

E	T	A	O	Z	Q

Which letter won? _____ Which was the rarest? _____

Key tracker

When you figure out a letter, write the cipher letter under the real letter. It helps you spot the rest!

Real	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Cipher																										

Flashcard mission

Make an index card with the 9 most common letters (E T A O I N S H R) and the one-letter words (A, I). Quiz yourself until you know them.

Case 4: Operation No Hint TOP SECRET

Open this case: after Lesson 4 (Fri Jan 29) · **Badge:** No-Hint Hero

> **Your mission:** this Aristocrat has **no hint**. Start with the one-letter words (A or I), then look for repeated words and letter patterns. No letter stands for itself.

1. No-hint Aristocrat (250 points)

N J M M L N H F M M N L B C B V G T N G X M

__ __

L N H H B G F M M

__ __ __ __ __ __ __ __ __

Detective notes

Which cipher letter is alone as a word? It must be A or I. Which cipher word shows up 3 times?

Letter guesses

Cipher	Real	Cipher	Real

Bonus: Vigenere (key: SPY)

Write the key under the message again and again (S P Y S P Y S P Y S…). Cipher number minus key number. Below 0? Add 26. Numbers are on page 4.

2. Vigenere (200 points)

E T C L P R F D M F

__ __ __ __ __ __ __ __ __ __

Key tracker

When you figure out a letter, write the cipher letter under the real letter. It helps you spot the rest!

Real	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Cipher																										

> **Memory race:** draw the Knight's Templar key, PigPen key, and Tap Code grid from memory 3 times each on page 11. Goal: each under 1 minute with 0 mistakes.

Case 5: Operation Final Countdown TOP SECRET

Open this case: after Lesson 5, the mock test (Fri Feb 26) · **Badge:** Speed Decoder

> **Your mission:** a timed mini test before the tournament. Set a timer for **15 minutes**. Do the questions you can finish fastest first. Then check with the answer key and score it: 2 or fewer mistakes = full points, each extra mistake loses 50.

1. Caesar (100 points) (Shift of 3 or less, either way.)

N V K R B Z M S T Q M S G D H Q G D Z C R

__ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __

2. Atbash (100 points)

H K R V H O L E V K F A A O V H

__ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __

3. Tap Code (150 points)

43 33 34 52 14 11 54

____ ____ ____ ____ ____ ____ ____

4. Baconian (150 points) (Sort the 4 symbols into an A group and a B group. Use the table on p. 5.)

Symbols	++*@*	*###+@	*+*##	*+@*+
A or B				
Letter				

5. PigPen (150 points)



____ ____ ____ ____ ____

Question	Mistakes	Points
Q1 Caesar		/ 100
Q2 Atbash		/ 100
Q3 Tap Code		/ 150
Q4 Baconian		/ 150
Q5 PigPen		/ 150
Total		/ 650

Time used: _____ minutes

Which 2 cipher types cost me the most points?

_____

_____

Redo that part of Lab I1, Lab I2, Lesson 3, or Lesson 4 before Sat Mar 13.

Memory Race

Cover page 5. Start a timer and draw each key from memory. Then check it with the key and write your time and mistakes. Get all 3 under **1 minute with 0 mistakes** to earn the **Memory Master** badge.

PigPen · Try 1

Date _____ Time _____ sec Mistakes ___

Tap Code · Try 1

Date _____ Time _____ sec Mistakes ___

Knight's Templar · Try 1

Date _____ Time _____ sec Mistakes ___

PigPen · Try 2

Date _____ Time _____ sec Mistakes ___

Tap Code · Try 2

Date _____ Time _____ sec Mistakes ___

Knight's Templar · Try 2

Date _____ Time _____ sec Mistakes ___

PigPen · Try 3

Date _____ Time _____ sec Mistakes ___

Tap Code · Try 3

Date _____ Time _____ sec Mistakes ___

Knight's Templar · Try 3

Date _____ Time _____ sec Mistakes ___

How to memorize a key: Draw the frame first, then write the letters in order in the spaces. Each letter's symbol is the shape of the lines around its space, plus a dot if its space has one. Draw it from memory once a day until it takes less than 1 minute.

Agent-to-Agent Cipher Swap

Real codebusters learn fast by **making** ciphers. Write 3 secret messages in 3 different cipher types (try Caesar, Atbash, PigPen, Tap Code, Vigenere with the key CAT, or Baconian). Fold the page so your partner can only see the cipher. Keep the real message on a sticky note so you can check.

Swap 1

Cipher type: _____ Key or shift: _____ Partner or family member: **decode it here**. Time: _____

My secret message (in cipher):

Swap 2

Cipher type: _____ Key or shift: _____ Partner or family member: **decode it here**. Time: _____

My secret message (in cipher):

Swap 3

Cipher type: _____ Key or shift: _____ Partner or family member: **decode it here**. Time: _____

My secret message (in cipher):

Tip: short messages of 3 to 6 words work best. Check your own cipher once before you swap it. 3 swaps earns the **Secret Writer** badge.

Evidence Log: my practice

Every practice counts: a case, a lab, a memory race, Cryptograms, or a practice test. Log 6 sessions to earn the Top Agent badge.

#	Date	Cipher type(s)	Minutes	Mistakes	Points or notes
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					
16					
17					
18					
19					
20					
21					
22					
23					
24					
25					

Case Missions

One case for each Codebusters meeting. Finish it, show a coach or your Team Captain for initials, then color the badge.

When	Mission	Badge	Done	Initials
After Fri Oct 2 Lesson 1	Case 1: crack 3 Caesars and 2 Atbash messages (p. 6).	Shift Cracker	☐	
After Fri Nov 6 Lesson 2	Case 2: Tap Code messages and the knock-knock challenge (p. 7).	Tap Talker	☐	
After Fri Nov 6 Lesson 2	Case 2: decode both PigPen messages (p. 7).	PigPen Pro	☐	
After Fri Dec 11 Lesson 3	Case 3: solve the hint Aristocrat and count letters (p. 8).	Letter Detective	☐	
After Fri Jan 29 Lesson 4	Case 4: solve the no-hint Aristocrat (p. 9).	No-Hint Hero	☐	
After Fri Feb 26 Lesson 5	Case 5: do and score the 15-minute mini test (p. 10).	Speed Decoder	☐	
Anytime Bonus	Memory race: all 3 keys under 1 minute (p. 11). Swap 3 ciphers (p. 12).	Memory Master / Secret Writer	☐	
All season Log	Log 6 practice sessions in your Evidence Log (p. 13).	Top Agent	☐	

Tournament day: Sat Mar 13

- ☐ Sharp pencils and an eraser
- ☐ A good night's sleep
- ☐ Draw all 3 memory keys one last time the night before

- Draw PigPen, Tap Code, and Templar on scrap paper first.
- Split the questions with your partner in the first minute.
- Do the fastest ones first. 2 small mistakes still get full points.

My Badges

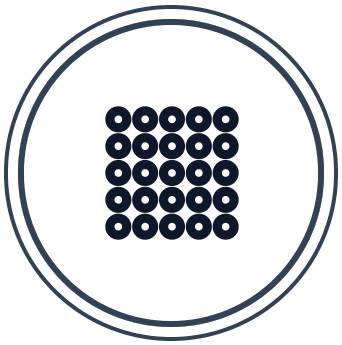
Color in each badge when you earn it. Earn all 9 to become a **Master Codebuster**! Your Team Badge Passport has the same badges.



Shift Cracker

Crack 3 Caesars and 2 Atbash (Case 1)

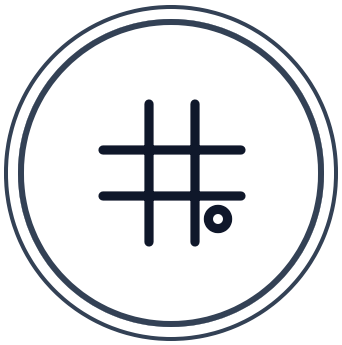
Initials: _____



Tap Talker

Tap and decode Tap Code (Case 2)

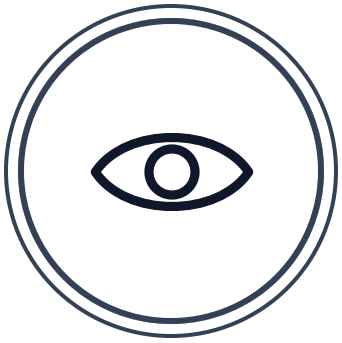
Initials: _____



PigPen Pro

Decode 2 PigPen messages (Case 2)

Initials: _____



Letter Detective

Solve a hint Aristocrat (Case 3)

Initials: _____



No-Hint Hero

Solve a no-hint Aristocrat (Case 4)

Initials: _____



Speed Decoder

Score the timed mini test (Case 5)

Initials: _____



Memory Master

Draw all 3 memory keys in under 1 minute

Initials: _____



Secret Writer

Write 3 ciphers for your partner

Initials: _____



Top Agent

Log 6 practice sessions

Initials: _____

Turn the book upside down to read the answers.

Answer key

Agents: finish the case first, then check. Fix mistakes in a different color.

Case 1: Operation Shift and Mirror

1) THE PASSWORD IS PANCAKE (made with shift +1, so move each letter back 1). 2) A SNAIL CAN SLEEP FOR THREE YEARS (made with shift -2, so move each letter forward 2). 3) HIDE THE MAP UNDER THE BIG ROCK (made with shift +3, so move each letter back 3). 4) FROGS DRINK WATER THROUGH THEIR SKIN. 5) THE CAKE IS IN THE FREEZER.

Case 2: Operation Knock-Knock

Tap: 1) MEET AT THE GATE. 2) WE DID IT. PigPen: 3) TOP SECRET. 4) RUN TO THE FORT.

Case 3: Operation Letter Count

THE BEST SPIES ARE GOOD AT LISTENING AND THEY NOTICE THE SMALL THINGS. The top cipher letters: H = E (8 times), R = T (8 times), D = S (6 times), Q = N (5 times), Z = I (5 times).

Case 4: Operation No Hint

1) A BEE CAN SEE A COLOR THAT WE CANNOT SEE. The lone letter is A; the word used 3 times is SEE. 2) MEET AT NOON.

Case 5: Operation Final Countdown

1) OWLS CAN TURN THEIR HEADS (made with shift -1). 2) SPIES LOVE PUZZLES. 3) SNOW DAY. 4) CODE (* and + are A; # and @ are B). 5) BRAVO. Total possible: 650 points.